

Developing an artificial neural network model to predict the durability of the RC beam by machine learning approaches

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ABSTRACT

The chloride diffusivity is a significant index to describe the degradation rate of concrete structures. At present, a lot of models have been proposed by previous studies to investigate the chloride diffusivity in concrete specimens. However, most of these models only consider the effect of exposure time, and erosion depth on the chloride diffusivity and ignore the uncertain factor in the environment. This may lead to a tremendous difference existed in the predicted-to-experimental ratios of all typical chloride diffusivity models. Machine learning approaches as potential alternative predictors are often used to cope with this bias. This paper adopted five contemporary ML approaches, namely, the backpropagation (BP) neural network, the decision tree (DT), the random forest (RF), the linear regression (LR), and ridge regression (RR) to explore the chloride diffusivity. Input features considered in this study are water-cement ratios, the thickness of concrete specimens, coarse aggregate fraction volume, ratios of environmental temperature to the maintenance standard temperature, and ratios of environmental humidity to the relative humidity. It was found that the backpropagation (BP) neural network is the most accurate prediction (85%) with the lowest root mean squared error. With this superiority, A study on the importance of the input parameters reveals that the volume fraction of coarse aggregate, water-cement ratios, the thickness of the protective layer, and ratios of environment humidity to the relative humidity are the most influential parameters on chloride diffusivity. In addition, a data-driven model was established to investigate the chloride diffusivity considering the effect of random factors both in material and environment. Compared with the test results, the results indicated that the prediction results are closer to the test results, which is helpful for in-depth exploring the durability of the concrete structures.

1. Introduction

Chloride-induced corrosion of reinforcing steel is one of the most important causes affecting the durability of reinforced concrete (RC) structures, impacting the economy and safety of RC structures. The chloride diffusivity as a significant index to describe the degradation rate of concrete structures should be in deep research, which is helpful to be useful in realistically predicting the durability of reinforced concrete structures [1–4].

Several studies make some contributions to estimate the chloride diffusivity of cementitious materials. For example, Thomas [5] investigated the chloride diffusivity and established a prediction model based on the second Fick's law. Wang [6–8] explored the effect of coarse aggregate volume fraction on chloride diffusivity and found that chloride diffusivity is negatively proportional to the coarse

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aggregate volume fraction. Lindvall [9] conducted a test to identify the effect of temperature on the chloride diffusivity of concrete and reported that chloride diffusivity is positively proportional to temperature. Choi [10] explored the chloride diffusivity by a test considering the coupling effect of water-cement and the exposure time. Finally, a theoretical model was proposed to predict chloride diffusivity. El Hassan et al. [11] demonstrated the effect of climatic conditions on chloride ingress in concrete. In addition, Han [12] et al. found that the thickness of the protective layer of the reinforced concrete structure increased, which is beneficial to reduce the chloride diffusivity of concrete structures.

Although a series of studies have investigated the chloride diffusivity in concrete, some models for predicting the chloride diffusivity have been developed. However, most of the previous models only consider the effect of random factors in material such as cement-water proportion, and coarse aggregate volume fraction on chloride diffusivity separately and ignore their coupling effect. Additionally, Conventional methods for establishing the relationship between these parameters are mainly based on statistical analysis [13–16], the accuracy of the regression results is determined by the number of test data and certain pre-defined linear or non-linear equations [17,18]. To counter these issues, large differences are often observed between the prediction results and the test solutions. Machine-learning techniques based on the concept of natural tools are developed to model complex engineering phenomena [19,20], which are usually used in various fields. For example, Yu [21] et al. investigated the shear strength of recycled concrete beams (RAC) based on the artificial neural network (ANN) model and the random forest (RF) model. The results show that the artificial neural network (ANN) model and the random forest (RF) model can predict the shear strength of recycled concrete beams (RAC) well. Kim [22] et al. explored the diffusion behavior of chloride ions in concrete structures by the neural network model and took the measured data of 30 concrete specimens as an input database to analyze the time-varying law of chloride diffusivity. T Gupta [23] utilized the adaptive artificial neural network ANN network to model the chloride ion permeability of concrete specimens, considering the effects of the environmental temperature. In this study, a data-driven model for the prediction of chloride diffusivity is developed based on machine learning approaches namely linear regression, ridge regression, decision tree, random forest, and backpropagation (BP) neural network, which are evaluated to examine the chloride diffusion estimation of concrete structures. Taking the water-cement ratios, the thickness of concrete specimens, coarse aggregate fraction volume, ratios of environmental temperature to the maintenance standard temperature, ratios of environmental humidity to critical relative humidity, and ratios of exposure time to maintenance time of specimens into consideration. Compared with previous models, it was found that the theoretical solutions obtained by the backpropagation (BP) neural network model are in good agreement with the test results, and the relative errors stay within 15%. In addition, this paper established a design model for chloride diffusivity based on the analyzed results of parameter importance and sensitivity.

2. Machine learning (ML) models

Machine learning (ML) models have been widely used in aerospace, hydrology, economics, civil engineering, and other fields including random forest, ridge regression, linear regression, neural network, decision tree, et al.

2.1. Backpropagation (BP) neural network model

The backpropagation (BP) model involves three layers: input layer, hidden layer, and output layer [24,25] The slope descent is used to set the thresholds and weights of the neurons in each layer, which is useful to cut down the error of the network's consequences until the error arrives at the preset convergence. an appropriate number of neurons in the hidden layer is obtained, which is good for reducing this model's mistake.

2.2. Decision tree model

The decision tree (DT) is a managed ML algorithm generating a tree-like model from training data. The probability of the approximated parameters' potential value is calculable by the decision tree (DT) model, even to assess this project's peril possibility. It is similar to a flowchart where each node stands for the examination of a feature and each branch denotes the test's result. The DT model's reply is obtainable, after adopting the decisions from the beginning node to the end node [26]. Relying on the splitting attribute, which is treated as an attribute room. R_m is segmented recursively. the attribution is a binary variable identifying that $a = 1 \{x_k > t\}$, for a threshold t , where x_k is either a numerical or binary attribute. To judge the objective production results, each concluding area is assigned a value. R_j was defined as the disjoint zones assigned to each leaf of the DT model, which was used to show the mapping connection between the input characteristics and output characteristics as demonstrated in Eq. (1). The default option hyper-parameters of Scikit-learn's decision tree role have been applied for building the DT model in this article.

$$g(x) = \sum_{j=1}^J b_j 1\{x \in R_j\} \quad (1)$$

2.3. Random forest model

The random forest (RF) is likewise a monitored ML approach, firstly recommended by Breiman, being composed of a combination of multiple decision trees, which is used to prevent the model's overfitting effectively. The tree classifiers are taken at random from the

given input attributes and created in varying sample training sets, which were used to build a decision tree by the bootstrap approach from the training dataset [27]. The random forest RF algorithm can be divided into two steps: the first step is to generate the RF model, and forecasting is executed in the second step which is based on the classifier generated in the first phase. There is a positive correlation between precision and the number of trees in the forest, and the most wonderful answer is accomplished through voting. The final model predicted output is taken by averaging individual decision trees' consequences by following Eq. (2).

$$Y = \frac{1}{n} \sum_{j=1}^n Z_b(M) \quad (2)$$

Where: Z_b means the individual decision trees, which are trained on M unknown instances. The symbol n denotes the total number of the decision trees in Eq. (2).

2.4. Linear regression

In statistics, linear regression (LR) is a managed machine that learns algorithm to determine a linear connection between the dependent and independent variables through the least square approach. In this circumstance, the regression coefficient can be treated as an important index to show the degree of linear connections between the independent and dependent variables [28]. The gradient descent is ordinarily the most effective option to take the regression coefficients. Relying on the given input variables, likewise realized as independent variables, the objective projection value is modeled on this algorithm. A hyperplane is fitted to the m -dimensional room where there is m number of independent attributes. The production (Y) represents the focused variable (chloride ions coefficient) in this algorithm model and can be written as follows.

$$Y = \beta_0 + \sum_{j=1}^m X_j \beta_j \quad (3)$$

X_j stands for the independent variable, β_0 means the Constant term, β_j denotes the regression coefficients ($j = 1, 2, \dots, m$).

2.5. Ridge regression model

Ridge regression is also realized as Gillonov regularization, corresponding to the Linear regression model, which is a biased estimation regression approach, that is dedicated to collinearity data analysis. By failing to take care of the partial information and giving up the least square method's unbiasedness, which is used to reduce the remaining total of squares. The coefficients are decreased in RR analysis as demonstrated in Eq. (3)[27].

$$\hat{\beta}_{ridge} = \operatorname{argmin}_{\beta} \sum_{i=1}^N \left[\left(y_i - \beta_0 - \sum_{j=1}^m X_{ij} \beta_j \right) + \lambda \sum_{j=1}^m \beta_j^2 \right]^2 \quad (4)$$

Here, the number of input attributes is denoted through m . While sample numbers in the dataset are denoted through i . The shrinkage parameter is symbolized by λ and the coefficients (β_j) tend to get close to zero while the value of λ grows.

3. Database establishment

According to the specification and experimental research, the thickness of the concrete protective layer, water-cement ratio, coarse aggregate volume fraction, ratios of exposure time to maintenance time, ratios of environmental temperature to the maintenance standard temperature, and ratios of environmental humidity to critical relative humidity were selected as random variables affecting the chloride diffusivity, as shown in Table 1. A total of 159 specimens were collected from researchers who study the performance of chloride diffusivity in concrete structures. When constructing the test database, the following criteria were applied:

- (1) This paper mainly investigated the chloride ions in plain concrete, data related to the influence of steel bars were excluded.
- (2) The chloride ions ingress concrete structures in different kinds of complex ways, such as convection and diffusion. Based on the previous studies [3,5], diffusion is the most important way for chloride ions transported into plain concrete that should pay

Table 1
Stochastic factors affecting the diffusion coefficient of chloride ions.

Stochastic factors	Symbols
Water-cement ratios	W/C
Coarse aggregate fraction volume	R
The thickness of the protective layer	M(cm)
Ratios of exposure time to maintenance time of specimens	t/t_0
Ratios of environmental temperature to maintenance standard temperature	T/T_0
Ratios of environment humidity to the relative humidity	h/h_c

more attention to. Therefore, the model established by the collected data is mainly used to predict the chloride diffusion coefficient of the diffusion zone in plain concrete.

(3) This paper only considers the chloride ions diffusion in plain concrete from one dimension.

The statistical results of the test parameters are shown in Fig. 1. [29–42].

Based on the database, the proportion of each parameter range is determined to show the influence of each parameter on the chloride diffusivity. Based on the study of Chen [43]. The proportion range of each parameter is shown in Fig. 2.

Fig. 2 indicated that the chloride diffusion coefficient is larger than 3×10^{-11} (83% specimens). While the coarse aggregate volume fraction of these concrete specimens greater than 0.45 only accounts for 20%, which is consistent with previous models that a larger coarse aggregate volume fraction can reduce the chloride ions diffusion coefficient. On the contrary, water-cement ratios, ratios of environmental temperature to maintenance standard temperature, and ratios of environmental humidity to the relative humidity increased will enhance the chloride diffusivity. Additionally, the database contains covers a wide range, which can well reflect the

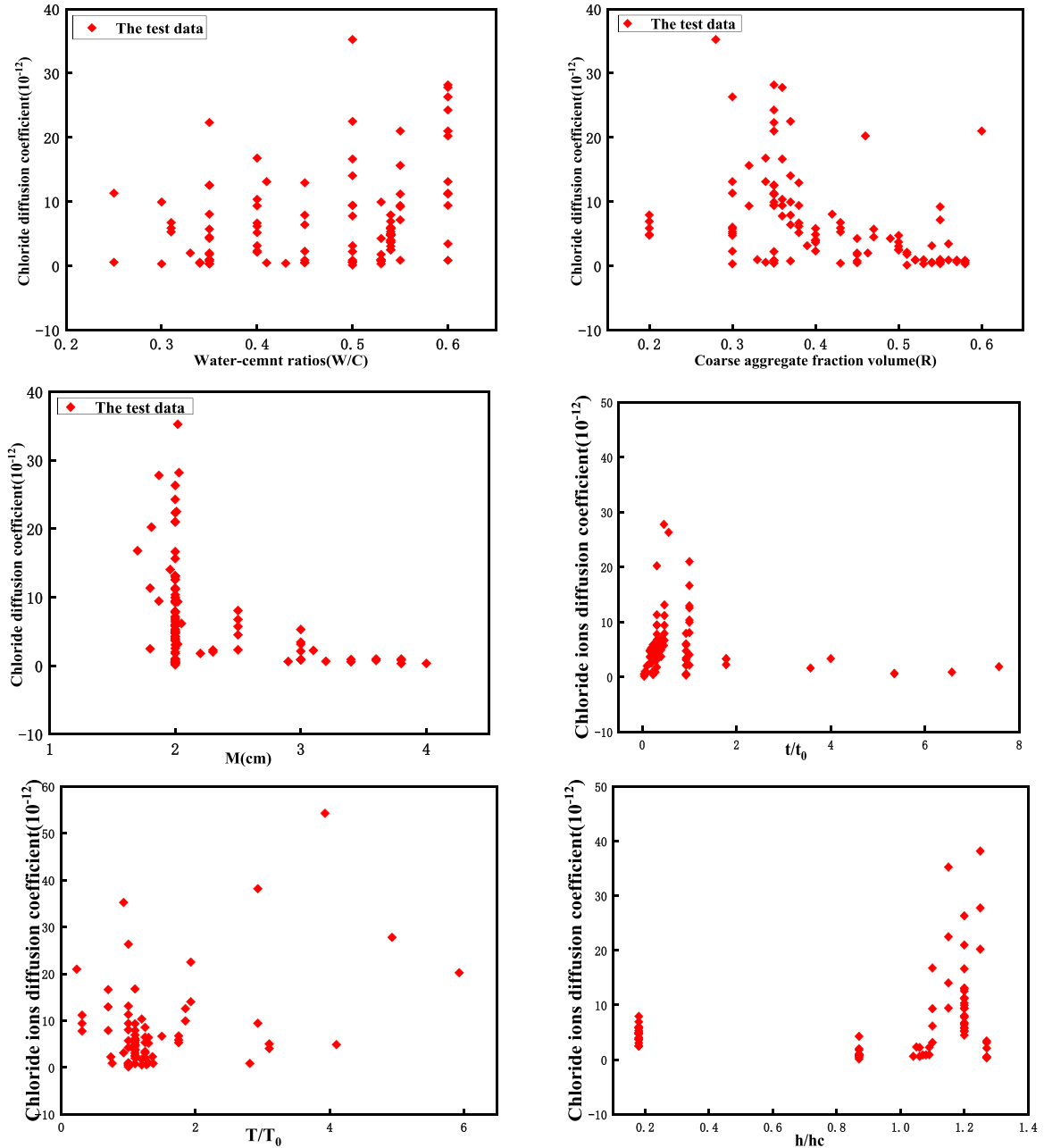
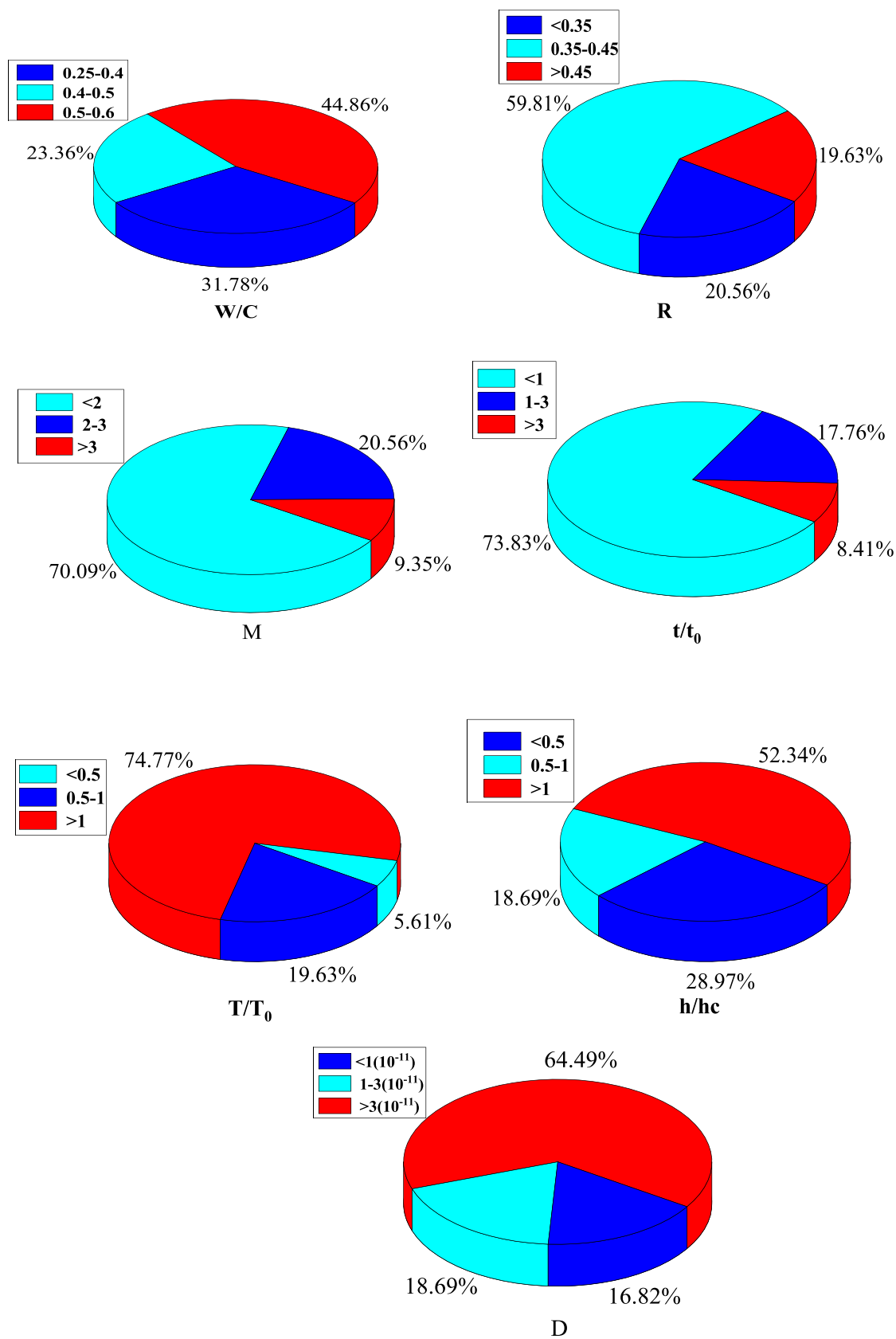


Fig. 1. Data source.



(caption on next page)

Fig. 2. Proportion of specimens.

influence on chloride diffusivity.

4. Evaluation of chloride diffusivity models recommended by researches

Based on experimental data and finite-element analyses of chloride ions diffusion diffusivity, many models have been proposed by researchers [2,5,6,10]. The theoretical models for predicting the chloride ions coefficient proposed by different researchers are shown in Table 2. Substituting the experimental data (as shown in Fig. 1) into these formulas (as shown in Table 2), the prediction results can be obtained. Compared with the test data, the comparison solutions are shown in Fig. 3.

D_0 means the chloride diffusivity when ($t = 28d$), m stands for the aging coefficient of the concrete material, V_p means the porosity of concrete, ρ_w is the density of water, ρ_c is the density of cement, is the degree of hydration, the value is 0.23, $(W/C)^*$ is water-to-cement ratio needed to attain complete hydration [44].

Fig. 3 illustrates the evaluation results of the diffusion diffusivity obtained by each model. the results show that the theoretical results obtained by the model proposed by Zheng J are too conservative and are much less than the experimental values. The relative errors between the test results and the theoretical solutions are greater than 15%, and the maximum error is over 30%. For the resident prediction models proposed by Wang [6], Thomas [5], Choi [10], there is also a big difference between the prediction results and the experiment solutions, the relative errors of each model are greater than 15%. Especially for the prediction model proposed by Chio, with the largest relative error (35%). The reason can be attributed to the fact that the previous models were developed based on a narrow range of data. Also, due to neglect the influence of some parameters on diffusion diffusivity.

5. Modeling of chloride ions coefficient of concrete structures

5.1. Prediction results of chloride ion concentration by various machine learning algorithms

From the above analysis, the results indicated that most of the previously proposed empirical equations for the prediction of chloride diffusivity of concrete structures are very conservative as they are developed based on limited data. This paper assessed several algorithms namely, linear regression (LR1), random forest (RF), ridge regression (RR), decision trees (DT), and back propagation BP neural network, which are established by Scikit-learn [45], as shown in Fig. 4. All the machine learning models are applied using a random test train split of the database. The training set and testing set are selected by using the random function in Scikit-learn package, which has reached a more objective and precise evaluation of the algorithms. The detailed steps can be expressed as follows: Firstly, the data were divided into several groups, and then the data were numbered by the random function. Finally, the training set and test set were selected by random sampling. The division is maintained at 60% (95samples) for the training subset and 40% (64 samples) for the testing subset. The prediction results of various mechanical algorithms are shown in Fig. 5. The comparison results between the test results and the prediction results are shown in Fig. 6.

From Fig. 6, the results show that the backpropagation (BP) neural network method is most suitable to simulate the chloride ions coefficient, the relative error results stay within 15%. And the random forest model is also showing a good performance, the decision tree is seen to give the poorest prediction.

5.2. Parametric study

This segment analyzes the sensitivity and significance of these parameters. The stages are as follows: while the defined approximation parameter's significance is studied, a advanced model removing it is built and then the advanced model's anticipated value is accessed. The contrasts between the coefficient of calculation of the advanced model and the original model are worked out. The coefficient of calculation between the predicted and the experimental values is calculable as follows.

Table 2
The previous theoretical model.

Previous models	Theoretical limitation model
Thomas [5]	$D = D_0 \left(\frac{t_0}{t}\right)^m m = 3 \times (0.55 - W/B)$
Wang [6]	$D = D_0 f(R), f(R) = 1 - 1.0182R$ $D_0 = 6.24 \times 10^{-12} m^2/s, m = 0.333$
Zheng J [2]	
Choi [10]	$D = \frac{2(1-R)}{2+R} \times D_0 \times 10^{-3.25\phi^2+7.25\phi-4}$ $\phi = \frac{W/C - \left[0.75k + 0.28\left(\frac{\rho_w}{\rho_c} + (W/C)^*\right)\right]k}{\frac{\rho_w}{\rho_c} + W/C}$ $k = 0.23, (W/C)^* = 0.38$

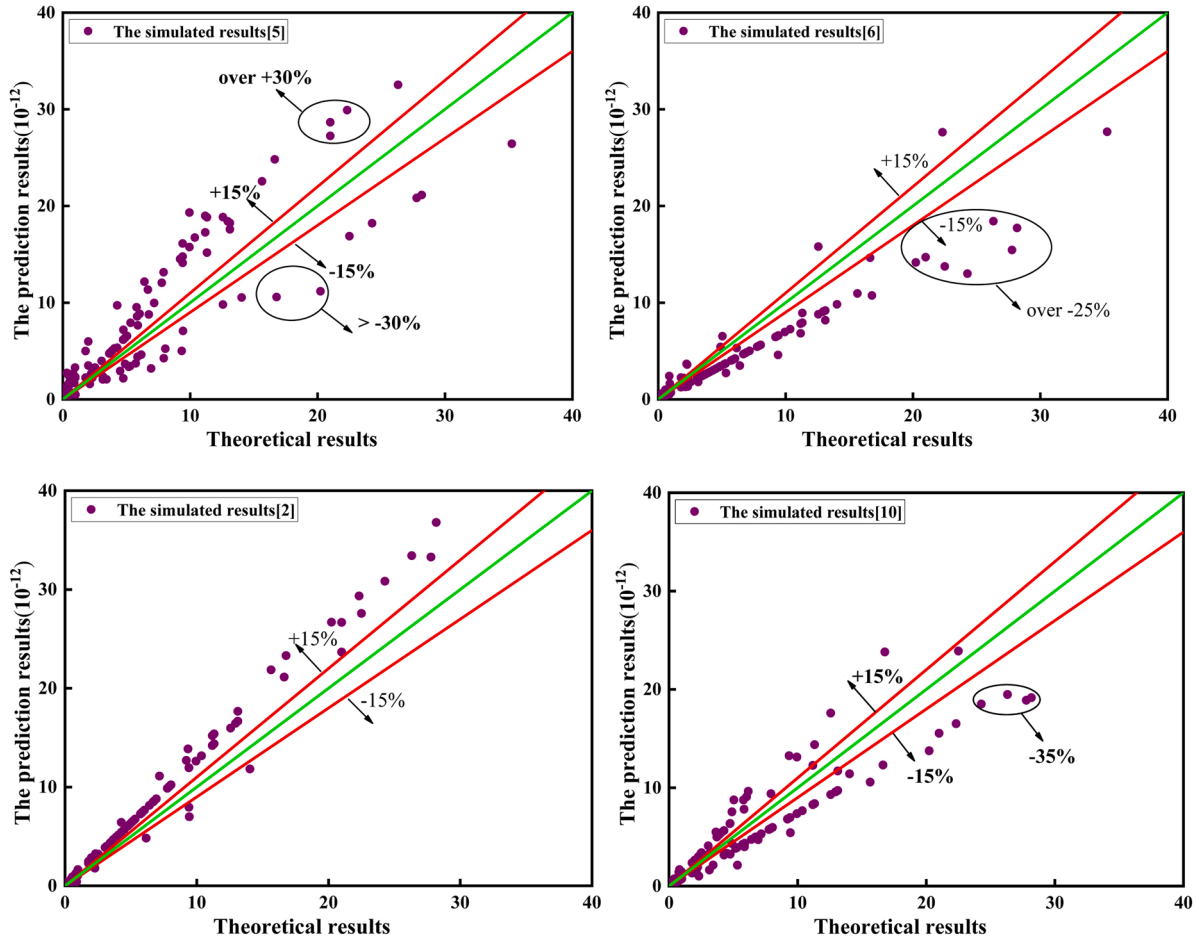


Fig. 3. Evaluation of the previous models.

$$Z^2 = \frac{\left[\sum_{i=1}^n (y - \bar{y})(y' - \bar{y}') \right]^2}{\sum_{i=1}^n (y - \bar{y})^2 \sum_{i=1}^n (y' - \bar{y}')^2} \quad (5)$$

Z^2 implies the coefficient of determination, where is the anticipated consequence of chloride ions coefficient, y is the real value, and n stands for the number of samples.

After getting rid of one parameter from this estimated model, the coefficient of determination enhanced, which indicates that this parameter owns a limited impact on prediction results. Table 3 and Fig. 7 show the importance analysis results of each parameter and the sensitivity analysis results respectively.

Table 3 shows that the R , W/C , M , and (h/h_c) will have a tremendous effect on the chloride diffusivity. Fig. 7 indicates the relationship between the chloride diffusivity and each parameter. The results show that chloride diffusivity is positively proportional to the (W/C) , (T/T_0) , (h/h_c) , and inversely proportional to the remaining parameters, which are consistent with the design models proposed by codes in 'Section 4'. This indicates the importance and sensitivity analysis results of the parameters obtained in this part are credible.

5.3. Development of a new model

Based on the above analysis, it was found that the chloride diffusivity in concrete structures is often affected by multiple factors, the value of chloride diffusivity will be enhanced, with the (W/C) , (T/T_0) , (h/h_c) increased, for the resident parameters, which will have a negative effect on it. To describe the nonlinear relationship between chloride ions coefficient and each parameter, a new model to predict chloride diffusivity was established by the nonlinear regression equation. This model includes seven parameters, the variables in the model are the power terms of each parameter as shown in Eq. (6).

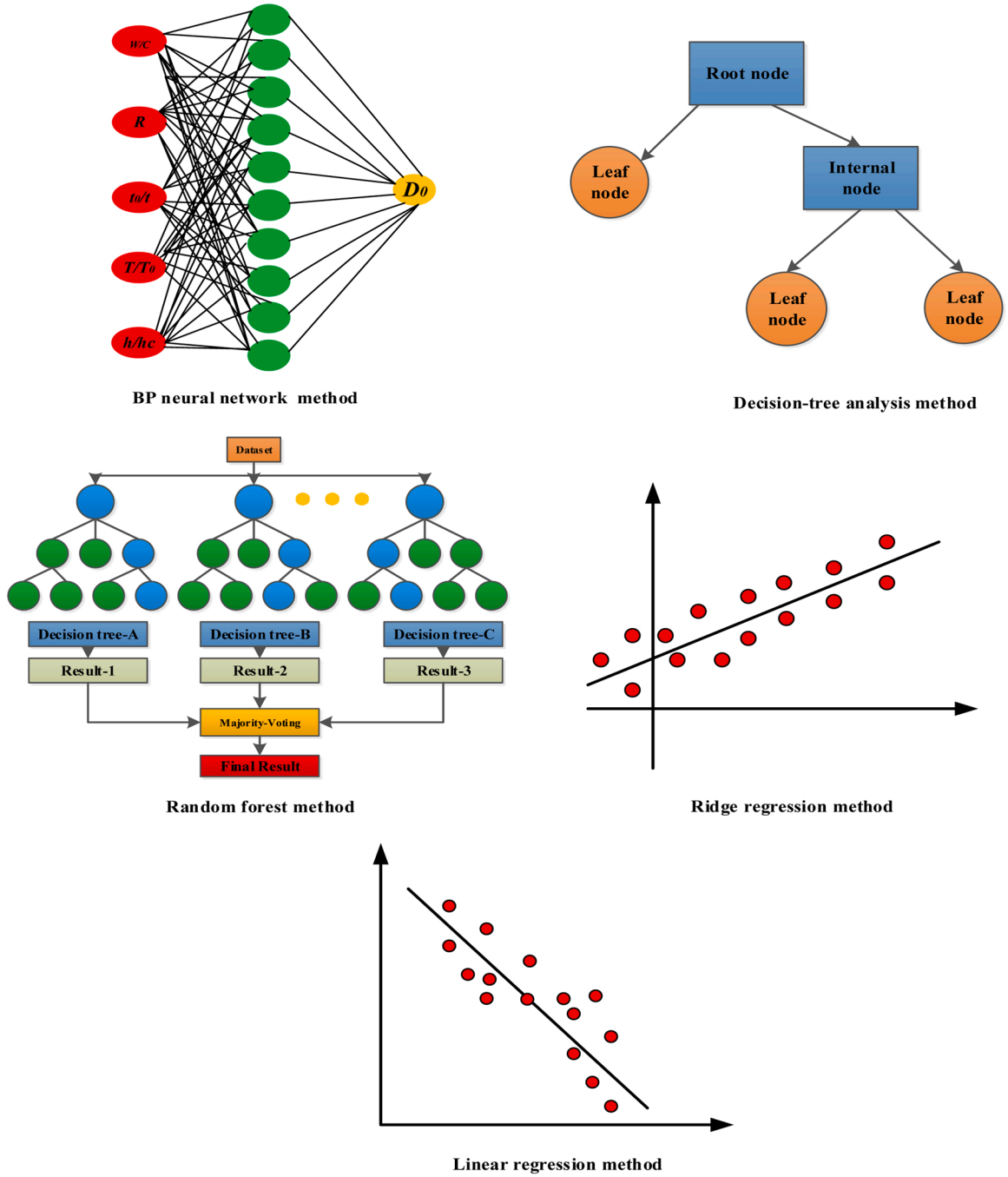


Fig. 4. Graphical representation of the ML algorithms.

$$D(x, t) = \alpha \times \left(\frac{W}{C}\right)^b \left(\frac{t_0}{t}\right)^c \left(\frac{T}{T_0}\right)^d \left(\frac{h}{h_c}\right)^e (R)^f (M)^Q \quad (6)$$

Taking both sides of Eq. (6) logarithmically, Eq. (6) can be rewritten as follows.

$$\ln D(x, t) = \ln \alpha + b \ln \left(\frac{W}{C}\right) + c \ln \left(\frac{t_0}{t}\right) + d \ln \left(\frac{T}{T_0}\right) + e \ln \left(\frac{h}{h_c}\right) + f \ln(R) + Q \ln(M) \quad (7)$$

The results of each parameter can be obtained as shown in Eq. (8).

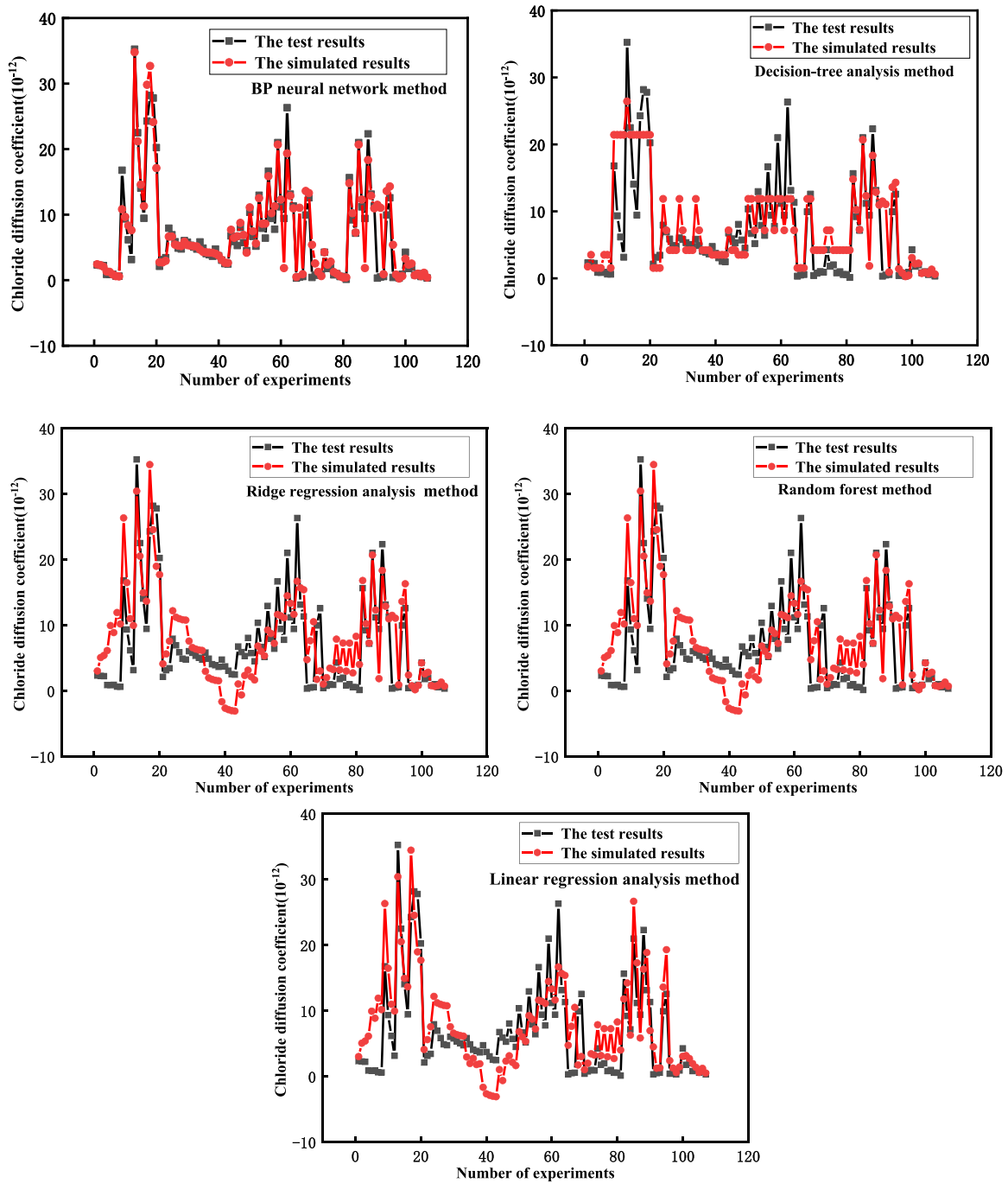


Fig. 5. The simulated results of chloride ions diffusion coefficient by machine learning algorithms.

$$D(x, t) = 13.96 \left(\frac{W}{C} \right)^{0.74} \left(\frac{t}{t_0} \right)^{-0.35} \left(\frac{T}{T_0} \right)^{0.31} \left(\frac{h}{h_c} \right)^{0.33} (R)^{-0.86} (M)^{-0.69} \quad (8)$$

From Eq. 8, it can be seen that the parameters that have a great influence on chloride diffusivity are coarse aggregate volume fraction, water-cement ratios, the thickness of the protective layer, and ratios of environment humidity to the relative humidity, which are consistent with the analysis results of sensitivity of each parameter as shown in Table 3 and Fig. 7 respectively.

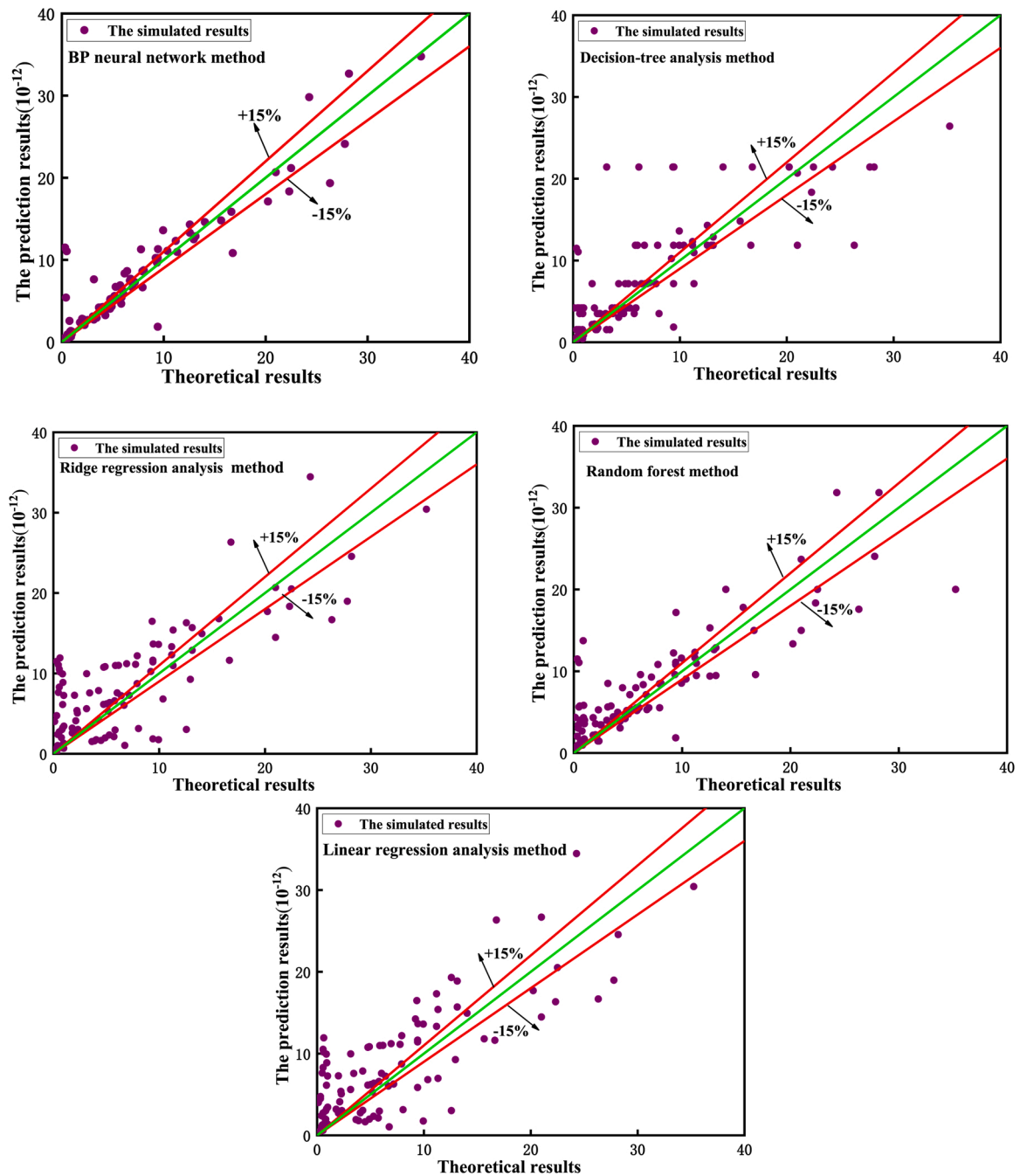


Fig. 6. The comparison results of the (ML) model and the experiment data.

Table 3
Importance of parameters.

All parameters	Coefficient of determination
W/C	0.38
R	0.33
M	0.47
t/t ₀	0.78
T/T ₀	0.73
h/hc	0.38

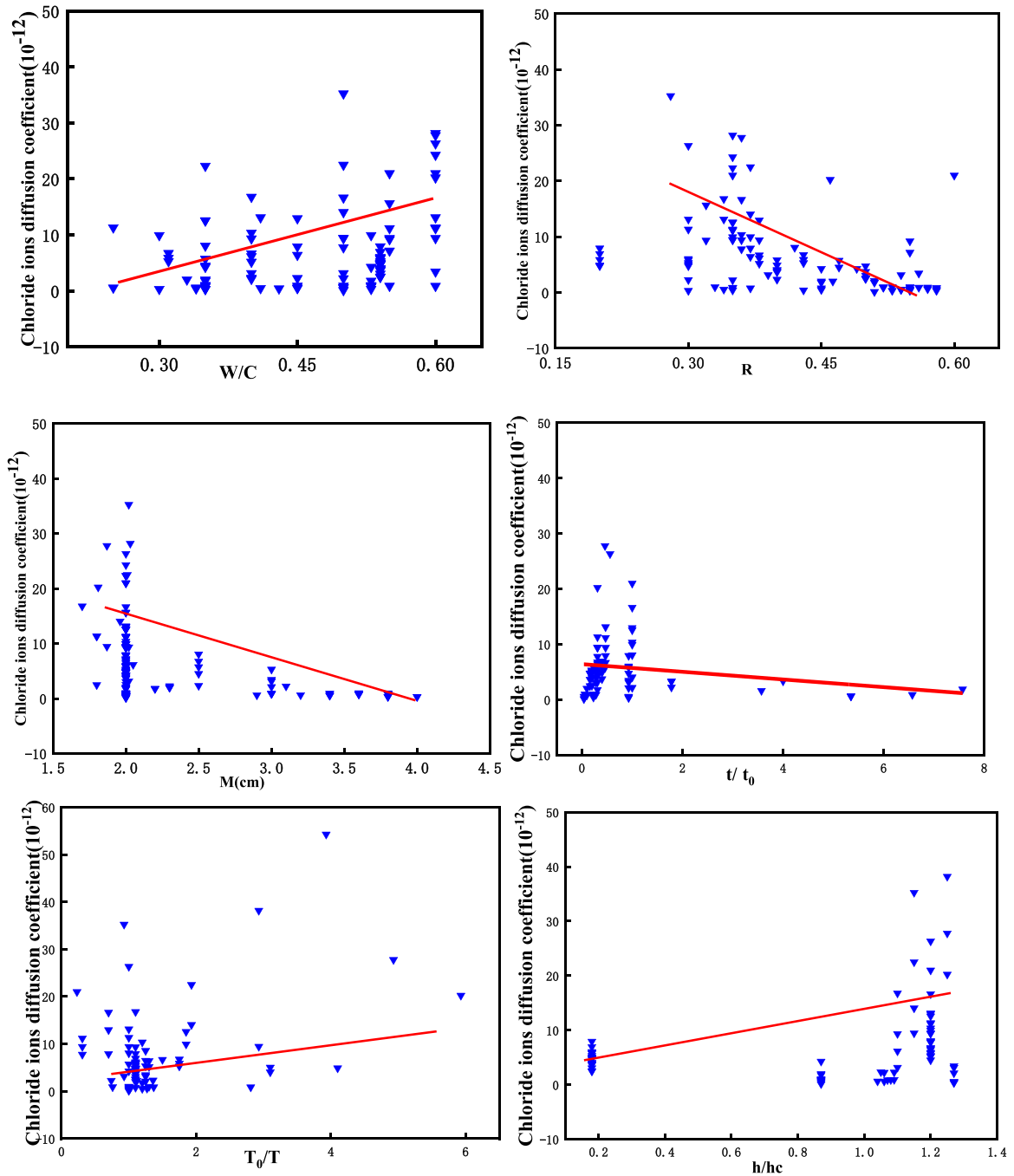


Fig. 7. Sensitivity of each parameter.

5.4. Model validation

To verify the calculation results of Eq. (8), the data in Fig. 1 were substituted into Eq. (8) and compared with the experimental data, the comparison results are shown in Fig. 8. From Fig. 8, it can be seen that the relative error between the prediction results is closer to the experimental data, the relative error stays within 15%, indicating that the model can predict the chloride ion diffusion coefficient well. Table 4 illustrated the evaluation of the researchers' model, and the model developed in this study.

As shown in Table 4, the average value and MAPE between the prediction results obtained by the backpropagation (BP) model and experimental values are 1.13 and 14.5% respectively.

Nevertheless, as for plenty of fitting algorithms, which can find a local minimal exclusively, but it cannot be ensured to be the worldwide minimum, this will lead to the

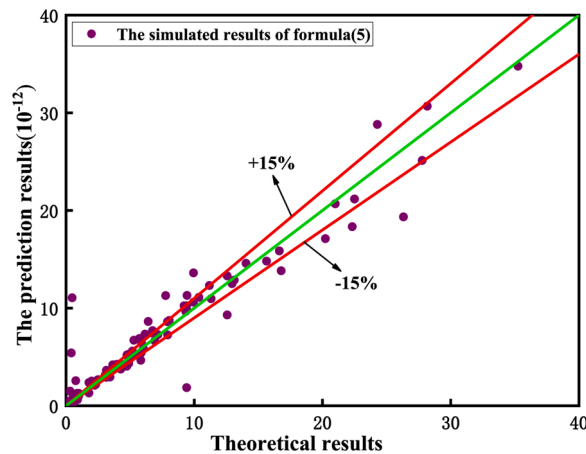


Fig. 8. The comparison results.

Table 4
Model evaluation.

Prediction model	Mean value	MAPE
Thomas [5]	1.28	28.40%
Wang [6]	0.71	36.90%
Zheng J [2]	1.35	43.50%
Choi [10]	0.81	29.80%
Proposed model	1.13	14.50%

parameters that were accessed in the purpose model not being the superlative values. To assess this model's reliability modernized in this research, a set of data was assembled, as demonstrated in Table 5. Substituting the collection data into formula(8), The prediction consequences, which are estimated by the BPNN model are obtained as demonstrated in Fig. 9.

As can be seen in Fig. 9, the MAPE of the six data sets evaluated by the proposed model in this study stays within 15%, which indicates that the model proposed by the BPNN can predict the chloride diffusivity well. However, this study just can be treated as an initial step to explore the chloride diffusivity of concrete structures, due to the limitation of test data. In future studies, the database should be expanded and more factors need to be considered to improve the accuracy of this prediction model.

6. Conclusion

This study presents 5 machine learning-based approaches to predict chloride diffusivity of concrete structures, based on the collected experimental data. The input parameters considered for the prediction of water-cement ratios, the thickness of concrete specimens, coarse aggregate fraction volume, ratios of environmental temperature to the maintenance standard temperature, ratios of environmental humidity to critical relative humidity, and ratios of environment humidity to the relative humidity. In addition, this study evaluated the previous theoretical models and developed a data-driven model based on the (ML) algorithm for predicting the chloride diffusivity. The conclusions can be obtained as follows.

- (1) A comparative study between experimental and predicted chloride diffusivity by previously proposed equations revealed that their predicted results have high variance and do not apply to a wide range of data.
- (2) The relative feature importance applied to the input features employed in the BPNN algorithm revealed that the most important feature in chloride diffusivity prediction is the coarse aggregate volume fraction, water-cement ratios, the thickness of the protective layer, and the relative humidity.
- (3) The simulated results with consideration of multiple factors based on the BPNN model are in good agreement with the test data. This illustrates the applicability of the developed data-driven model and that the environmental factors such as environmental temperature and humidity should be taken into account in this model.

Data availability

All data, models, and code generated or used during the study appear in the submitted article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

Table 5
Data set.

	Specimens	W/B	R	M (cm)	t/t_0	T/T_0	h/h_c
[10]	C40	0.4	0.48	5.00	0.107	0.83	1.26
		0.4	0.48	5.01	0.25	0.83	1.26
		0.4	0.48	5.05	1	0.83	1.26
		0.4	0.48	4.86	3.25	0.83	1.26
	C50	0.5	0.47	5.02	0.107	0.81	1.26
		0.5	0.47	5.03	0.25	0.81	1.26
		0.5	0.47	5.03	1	0.81	1.26
		0.5	0.47	5.03	3.25	0.81	1.26
	C60	0.6	0.46	5.03	0.107	0.83	1.26
		0.6	0.46	5.03	0.25	0.83	1.26
		0.6	0.46	5.03	1	0.83	1.26
		0.6	0.46	5.03	3.25	0.83	1.26

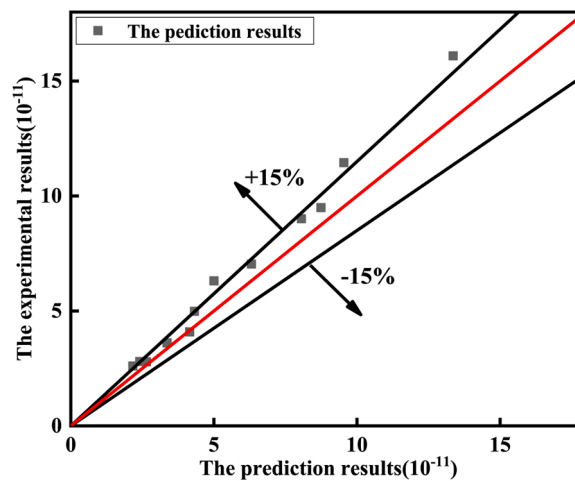


Fig. 9. Model evaluation.

influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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References

- [1] N.D. Hoang, C.T. Chen, K.W. Liao, Prediction of chloride diffusion in cement mortar using Multi-Gene Genetic Programming and Multivariate Adaptive Regression Splines[J], Measurement (2017) 141–149.
- [2] J. Zheng, X. Zhou, Analytical solution for the chloride diffusivity of hardened cement paste, J. Mater. Civ. Eng. 10 (2008) 1061.
- [3] H. Farahani, M. Shekarchi Taghaddos, Prediction of long-term chloride diffusion in silica fume concrete in a marine environment, Cem. Concr. Compos. 59 (2015) 10–17.
- [4] L. Pang, Q. Li, Service life prediction of RC structures in the marine environment using long term chloride ingress data: comparison between exposure trials and real structure surveys, Constr. Build. Mater. 11 (3) (2016) 979–987.
- [5] M.D.A. Thomas, P.B. Bamforth, Modelling chloride diffusion in concrete effect of fly ash and slag, Cem. Concr. Res. 29 (4) (1999) 487–495.
- [6] Y. Wang, X. Gong, L.J. Wu, Prediction model of chloride diffusion in concrete considering the coupling effects of coarse aggregate and steel reinforcement exposed to marine tidal environment[J], Constr. Build. Mater. 216 (20) (2019) 40–57.
- [7] Y.Z. Wang, C.X. Liu, Y.C. Wang, Q.M. Li, H. Liu, Time-and-depth-dependent model of chloride diffusion coefficient for concrete members considering the effect of coarse aggregate, J. Mater. Civ. Eng. 30 (3) (2018) 1–12.
- [8] Z.Q. Jin, X. Zhao, T.J. Zhao, J.Q. Li, Chloride ions transportation behavior and binding capacity of concrete exposed to different marine corrosion zones, Constr. Build. Mater. 177 (2018) 170–183.
- [9] A. Lindvall, Chloride ingress data from field and laboratory exposure influence of salinity and temperature, Cem. Concr. Compos. 29 (2007) 88–93.
- [10] Y.C. Choi, B. Park, G.S. Pang, et al., Modelling of chloride diffusivity in concrete considering effect of aggregates[J], Constr. Build. Mater. 136 (1) (2017) 81–87.

- [11] El Hassan, P. Bressolette, A. Chateauf, El TawilK, Reliability-based assessment of the effect of climatic conditions on the corrosion of RC structures subject to chloride ingress, *Eng. Struct.* 32 (10) (2010) 3279–3287.
- [12] Han Yanqing, Analysis on Thickness Control and Durability of Protective Layer of Coastal Concrete Structure [D], Qingdao University of Technology, 2012.
- [13] Q. Liu, Z. Hu, X. Lu, J. Yang, I. Azim, W. Sun, Prediction of chloride distribution for offshore concrete based on statistical analysis, *Materials* 13 (2020) 174–179.
- [14] A. Silva, R. Neves, J. de Brito, Statistical modelling of the influential factors on chloride penetration in concrete, *Cem. Concr. Res.* 69 (5) (2017) 255–270.
- [15] A.H. Gandomi, A.H. Alavi, M.R. Mirzahassemi, F.M. Nejad, Nonlinear genetic-based models for prediction of flow number of asphalt mixtures, *J. Mater. Civ. Eng.* 3 (23) (2011) 248–263.
- [16] L.Y. Li, J. Xia, S.S. Lin, A multi-phase model for predicting the effective diffusion coefficient of chlorides in concrete, *Constr. Build. Mater.* 1 (26) (2012) 295–301.
- [17] S. Caré, Influence of aggregates on chloride diffusion coefficient into mortar, *Cem. Concr. Res.* 11 (7) (2003) 1021–1102.
- [18] A. Delagrave, J.P. Bigas, J.P. Olivier, J. Marchand, M. Pigeon, Influence of the interfacial zone on the chloride diffusivity of mortars, *Adv. Cem. Based Mater.* 2 (5) (1997) 86–92.
- [19] J.J. Xu, X.Y. Zhao, Y. Yu, et al., Parametric sensitivity analysis and modelling of mechanical properties of normal-and high-strength recycled aggregate concrete using grey theory, multiple nonlinear regression and artificial neural networks, *Constr. Build. Mater.* 211 (2019) 479–491.
- [20] Y. Yu, X.Y. Zhao, J.J. Xu, et al., Evaluation of shear capacity of steel fiber reinforced concrete beams without stirrups using artificial intelligence, *Materials* 15 (2022) 2407.
- [21] M.A. DeRousseau, J.R. Kasprzyk, W.V. Srubar III, Computational design optimization of concrete mixtures: a review, *Cem. Concr. Res.* 109 (04) (2018) 42–53.
- [22] D. Kim, H. Lee, S. Lee, et al., A study on the evaluation of probabilistic durability life for RC structures deteriorated by chloride ion[J], *Key Eng. Mater.* 348–349 (2007) 417–420.
- [23] T. Gupta, K.A. Patel, S. Siddique, R.K. Sharma, S. Chaudhary, Prediction of mechanical properties of rubberised concrete exposed to elevated temperature using ANN, *Measurement* 147 (05) (2019), 106870.
- [24] A. Oztas, M. Pala, E. Özbay, E. Kancan, N. Çalar, M.A. Bhatti, Predicting the compressive strength and slump of high strength concrete using neural network, *Constr. Build. Mater.* 20 (2) (2006) 769–775.
- [25] A.T.A. Dantas, M. Batista, K. Leite, Nagahama de Jesus, Prediction of compressive strength of concrete containing construction and demolition waste using artificial neural networks, *Constr. Build. Mater.* 38 (02) (2013) 717–722.
- [26] S. Kameshwar, S. Misra, J.E. Padgett, Decision tree-based bridge restoration models for extreme event performance assessment of regional road networks, *Struct. Infrastruct. Eng.* 16 (3) (2020) 431–451.
- [27] L. Breiman, Random forests, *Mach. Learn.* 45 (1) (2001) 5–32.
- [28] H. Liang, W. Song, Improved estimation in multiple linear regression models with measurement error and general constraint, *J. Multivar. Anal.* 100 (4) (2009) 726–741.
- [29] Y. Yan, R. Liu, C. Lu, et al., Impact of curing humidity on chloride transportation in concrete[J], *J. Harbin Inst. Technol.* (12) (2016) 21–37.
- [30] Yun Zhang, Xiaoyun Zhou, Meng Wang, et al., Differences between time-dependent instantaneous and apparent chloride diffusion coefficients of concrete in tidal environment, *Constr. Build. Mater.* (10) (2020) 1943–5533.
- [31] A. Delagrave, J.P. Bigas, J.P. Ollivier, et al., Influence of the interfacial zone on the chloride diffusivity of mortars[J], *Constr. Build. Mater.* 5 (3–4) (1997) 86–92.
- [32] Hu Shouwang, Jianxin, et al., Influences of time, temperature, and humidity on chloride diffusivity: mesoscopic numerical research[J], *J. Mater. Civ. Eng.* 29 (11) (2017) 1943–5533.
- [33] C.C. Yang, J.K. Su, Approximate migration coefficient of interfacial transition zone and the effect of aggregate content on the migration coefficient of mortar, *Cem. Concr. Res.* 10 (2002) 1559–1565.
- [34] R.A. De Medeiros-Junior, P.C. de Brito M G Lima, M H F de Medeiros, Chloride penetration into concrete in an offshore platform-analysis of exposure conditions, *Ocean Eng.* 07 (2015) 78–87.
- [35] K. Audenaert, Q. Yuan, G. De Schutter, On the time dependency of the chloride migration coefficient in concrete, *Constr. Build. Mater.* 3 (2010) 396–402.
- [36] M.N. Chari, M. Shekarchi, M.H. Tadayon, M. Moradian, Prediction of chloride ingress into blended cement concrete: Evaluation of a combined short-term laboratory-numerical procedure, *Constr. Build. Mater.* 06 (2018) 649–662.
- [37] A. Chauhan, U.K. Sharma, Influence of temperature and relative humidity variations on non-uniform corrosion of reinforced concrete, *Structures* 07 (2019) 296–308.
- [38] S.W. Pack, M.S. Jung, H.W. Song, S.H. Kim, K.Y. Ann, Prediction of time dependent chloride transport in concrete structures exposed to a marine environment, *Cem. Concr. Res.* 2 (2010) 302–312.
- [39] J. Peng, Z. Li, B. Ma, Neural network analysis of chloride diffusion in concrete, *J. Mater. Civ. Eng.* 14 (4) (2002) 327–333.
- [40] B. Qifa, C. Mfia, Y. Jian, et al., Prediction of chloride diffusivity in concrete using artificial neural network: modelling and performance evaluation ScienceDirect [J], *Constr. Build. Mater.* 134 (02) (2020) 268–278.
- [41] B. Bourdette, E. Ringot, J.P. Ollivier, Modelling of the transition zone porosity, *Cem. Concr. Res.* 25 (1995) 741–751.
- [42] K. Zhao, Y. Qiao, P. Zhang, J. Bao, Y. Tian, Experimental and numerical study on chloride transport in cement mortar during the drying process, *Constr. Build. Mater.* 258 (05) (2020), 119655.
- [43] C. Dan, Research on the Decline Law of Shear Bearing Capacity of Reinforced Concrete Beams in Offshore Environment [D], Chong Qing Jiao Tong University, 2019.
- [44] G.S. Pang, S.T. Chae, S.P. Chang, Predicting model for pore structure of concrete including interface transition zone between aggregate and cement paste, *Int. J. Concr. Struct.* 3 (2009) 81–90.
- [45] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, et al., Machine learning in Python, *J. Mach. Learn. Res.* 12 (2011) 2825–2830.